

Benjamin (Min) Wesley Priest (they/them)

CONTACT INFORMATION

Computing Scientist
Center for Applied Scientific Computing
Lawrence Livermore National Laboratory
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E-mail: priest2@llnl.gov

RESEARCH INTERESTS

Efficient analysis of large, dynamic datasets: sketching, streaming algorithms, machine learning, high performance computing, graph algorithms, numerical linear algebra, compressed sensing, graph theory, optimization, network analysis, and theory of deep learning.

RESEARCH EXPERIENCE

Lawrence Livermore National Laboratory, Livermore, CA, USA

Center for Applied Scientific Computing, Supervisor: Dr. Kathryn Mohror

Graph Science and Irregular Computing Group Leader

10/2025 – present

First line manager of a team of 11 computer scientists and applied mathematicians whose work includes the full scope of graph, hypergraph, and network science from theory to advanced software as well as the development of state-of-the-art high performance computing software for managing the irregular communication workloads that arise in graph and data science workloads. Responsibilities include advising staff on career planning, connecting them with funding opportunities, and evaluating their performance for the division. Additional responsibilities include working with the Computing Directorate's Workforce Division on culture improvement initiatives. These include the Thrive Conversations – a campaign every six months involving interviewing a ~100-person sample of the ~1500 Computing employees to identify workforce attitudes towards the Lab culture and any pain points, which we use to deliver anonymized, aggregate guidance to Computing directorate leadership. I am an inaugural member of the Thrive Conversation Committee and have been involved since its inception in 2024. I am also involved in initiatives intending to address concerns raised in the Thrive Conversations, such as the incipient Stress and Burnout Initiative intended to understand and ameliorate sources of stress among the Computing workforce.

Computing Scientist

02/2021 – present

PI and Co-I of multiple research projects investigating scalable graph analytics, machine learning, and statistical modeling on High-Performance Computing (HPC) systems. Project funding sources include LLNL Laboratory Directed Research and Development, DOE Office of Science Advanced Scientific Computing Research, and U.S. Government Agency sponsors. Supervised 3 postdocs and dozens of graduate students. Selected research contributions include distributed subspace embedding and sketches [1, 2], distributed K nearest neighbors [3, 4, 5, 6], distributed power iteration of large structured matrices [7], distributed clustering of large and high-dimensional data [8], communication infrastructure for asynchronously handling the irregular communication patterns endemic to graph algorithms and related computation [9, 10], novel algorithms and software for scalable Gaussian process (GP) estimation [11, 12], and cosmology, climate, and space domain modeling [13].

Data Science Summer Internship Lead

10/2024 – present

Manager and facilitator of the **Data Science Summer Internship**, a yearly summer program serving ~40 students and matching LLNL staff mentors each year. Responsibilities include managing intern participation in training and networking events such as the **Data Science Challenge** and the **CASIS Annual Workshop**, working with mentors to scope and approve summer projects for each student, managing relationships with external partners such as Japan Science and Technology Agency and Japan Agency for Medical Research and Development, and planning and implementing staff- and professor-taught short courses and seminars that are broadcast to the whole Lab population.

Postdoctoral Researcher

04/2019 – 02/2021

Developed novel sketching algorithms to cluster [1] and perform local query approximation [14] massive graphs on HPC. Solved reinforcement learning [15], image classification [16], and quantum machine learning [17] problems using Gaussian Processes (GPs) and neural kernels. Created an HPC library implementing scalable GP algorithms [12].

Computation Student Intern **05/2018 – 01/2019**
 Co-created novel communication library for distributed problems with irregular communication patterns, such as sparse linear algebra and graph problems [18, 10]. Used cardinality sketches to estimate local triangle counts in distributed graphs [19].

Dartmouth College, Hanover, NH, USA

Thayer School of Engineering. Advisor: Professor George Cybenko

Research and Teaching Assistant **09/2015 – 02/2019**
 Invented streaming approximation algorithms for several centrality indices on massive graphs using sketches. Designed game and graph-theoretic models for advanced persistent threats in cyber defense. Taught courses in machine learning and lead a team of TAs.

MIT Lincoln Laboratory, Lexington, MA, USA

Cyber Analytics and Decision Systems. Supervisor: Dr. Kevin M. Carter

Assistant Research Scientist **08/2011 – 07/2015**
 Modeled computer networks using novel machine learning algorithms. Developed multi-agent systems for high-fidelity network simulations and cyber defense evaluation.

Air Force Institute of Technology, Wright-Patterson Air Force Base, OH, USA

Program Encryption Group. Supervisor: Professor J. Todd McDonald

Engineering Technician GS-05 **Summer, 2008 & 2009**
 Developed encryption metrics for circuits using abstract interpretation semantic models.

EDUCATION

Thayer School of Engineering at Dartmouth College, Hanover, VT, USA

Ph.D., Engineering **09/2015 – 02/2019**
 - Advisor: Professor George Cybenko
 - Thesis: Sublinear Approximations of Vertex Centrality in Evolving Graphs

The Ohio State University, Columbus, OH, USA

B.S., Mathematics **09/2007 – 06/2011**
 B.S., Computer and Information Science **09/2007 – 06/2011**

PEER-REVIEWED CONFERENCE PUBLICATIONS

- [20] Nate Veldt, Thomas Stanley, **Benjamin W Priest**, Trevor Steil, Keita Iwabuchi, TS Jayram, Grace J Li, and Geoffrey Sanders. Better learning-augmented spanning tree algorithms via metric forest completion. In *The Fourteenth International Conference on Learning Representations*
- [21] Nate Veldt, Thomas Stanley, **Benjamin W Priest**, Trevor Steil, Keita Iwabuchi, T. S. Jayram, and Geoffrey Sanders. Approximate forest completion and learning-augmented algorithms for metric minimum spanning trees. In *Proceedings The International Conference on Machine Learning (ICML)*, 2025
- [4] Keita Iwabuchi, Trevor Steil, **Benjamin W Priest**, Roger Pearce, and Geoffrey Sanders. NEO-DNND: Communication-optimized distributed nearest neighbor graph construction. In *Proceedings of the SC'24 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis*, 2024
- [9] Trevor Steil, Tahsin Reza, **Benjamin W Priest**, and Roger Pearce. Embracing irregular parallelism in HPC with YGM. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–13, 2023
- [5] Keita Iwabuchi, Trevor Steil, **Benjamin W Priest**, Roger Pearce, and Geoffrey Sanders. Towards a massive-scale distributed neighborhood graph construction. In *Proceedings of the SC'23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis*, pages 728–738, 2023

- [22] Imène R Goumiri, Amanda L Muyskens, **Benjamin W Priest**, and Robert E Armstrong. Light curve forecasting and anomaly detection using scalable, anisotropic, and heteroscedastic Gaussian process models. In *Proceedings of the Advanced Maui Optics and Space Surveillance (AMOS) Conference*, 2023
- [23] Rafael Bidese, Chinedu Eleh, Yunli Zhang, Roberto Molinari, Nedret Billor, **Benjamin W Priest**, Imène R Goumiri, Amanda L Muyskens, and Alec M Dunton. Stellar blend image classification using computationally efficient Gaussian processes (MuyGPs). *arXiv preprint arXiv:2208.14592*, 2022
- [24] Killian Wood, Alec M Dunton, Amanda Muyskens, and **Benjamin W Priest**. Scalable Gaussian process hyperparameter optimization via coverage regularization. *arXiv preprint arXiv:2209.11280*, 2022
- [25] Imène R Goumiri, Alec M Dunton, Amanda L Muyskens, **Benjamin W Priest**, and Robert E Armstrong. Light curve completion and forecasting using fast and scalable Gaussian processes (MuyGPs). In *Proceedings of the Advanced Maui Optics and Space Surveillance (AMOS) Conference*, 2022. *arXiv preprint arXiv:2208.14592*
- [26] Alec M Dunton, **Benjamin W Priest**, and Amanda Muyskens. Fast Gaussian process posterior mean prediction via local cross validation and precomputation. *arXiv preprint arXiv:2205.10879*, 2022
- [27] Trevor Steil, Tahsin Reza, Keita Iwabuchi, **Benjamin W Priest**, Geoffrey Sanders, and Roger Pearce. TriPoll: computing surveys of triangles in massive-scale temporal graphs with metadata. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–12, 2021
- [16] Imène R Goumiri, Amanda L Muyskens, Michael D Schneider, **Benjamin W Priest**, and Robert E Armstrong. Star-galaxy separation via Gaussian processes with model reduction. In *2020 Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS)*, 2020
- [1] **Benjamin W Priest**, Alec Dunton, and Geoffrey Sanders. Scaling graph clustering with distributed sketches. In *2020 IEEE High Performance Extreme Computing Conference (HPEC)*, pages 1–7. IEEE, 2020
- [15] Imène R Goumiri, **Benjamin W Priest**, and Michael D Schneider. Reinforcement learning via Gaussian processes with neural network dual kernels. In *2020 IEEE Conference on Games (CoG)*, pages 1–8. IEEE, 2020
- [28] Trevor Steil, Scott McMillan, Geoffrey Sanders, Roger Pearce, and **Benjamin W Priest**. Kronecker graph generation with ground truth for 4-cycles and dense structure in bipartite graphs. In *2020 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, pages 237–246. IEEE, 2020
- [29] Roger Pearce, Trevor Steil, **Benjamin W Priest**, and Geoffrey Sanders. One quadrillion triangles queried on one million processors. In *2019 IEEE High Performance Extreme Computing Conference (HPEC)*, pages 1–5. IEEE, 2019
- [18] **Benjamin W Priest**, Trevor Steil, Geoffrey Sanders, and Roger Pearce. You’ve got mail (ygm): Building missing asynchronous communication primitives. In *2019 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, pages 221–230. IEEE, 2019
- [30] Trevor Steil, **Benjamin W Priest**, Geoffrey Sanders, Roger Pearce, Timothy La Fond, and Keita Iwabuchi. Distributed kronecker graph generation with ground truth of many graph properties. In *2019 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, pages 251–260. IEEE, 2019

- [19] **Benjamin W Priest**, Roger Pearce, and Geoffrey Sanders. Estimating edge-local triangle count heavy hitters in edge-linear time and almost-vertex-linear space. In *2018 IEEE High Performance extreme Computing Conference (HPEC)*, pages 1–7. IEEE, 2018
- [31] Luan Huy Pham, Massimiliano Albanese, and **Benjamin W Priest**. A quantitative framework to model advanced persistent threats. In *ICETE (2)*, pages 448–459, 2018
- [32] **Benjamin W Priest**, Era Vuksani, Neal Wagner, Brady Tello, Kevin M Carter, and William W Streilein. Agent-based simulation in support of moving target cyber defense technology development and evaluation. In *Proceedings of the 18th Symposium on Communications & Networking*, pages 16–23, 2015
- [33] Kevin M Carter, Rajmonda S Caceres, and **Benjamin W Priest**. Characterization of latent social networks discovered through computer network logs. In *Networks in the Social and Information Sciences workshop of the 29th Annual Conference on Neural Information Processing Systems*, 2015
- [34] **Benjamin W Priest** and Kevin M Carter. Characterizing latent user interests on enterprise networks. In *The Twenty-Seventh International Flairs Conference*, 2014
- [35] Kevin M Carter, Rajmonda S Caceres, and **Benjamin W Priest**. Latent community discovery through enterprise user search query modeling. In *Proceedings of the 37th international ACM SIGIR conference on Research & development in information retrieval*, pages 871–874, 2014
- [36] Kevin Gold, Zachary J Weber, **Benjamin W Priest**, Josh Ziegler, Karen Sittig, William W Streilein, and Mark Mazumder. Modeling how thinking about the past and future impacts network traffic with the gosmr architecture. In *Proceedings of the 2013 international conference on Autonomous agents and multi-agent systems*, pages 127–134. Citeseer, 2013
- [37] **Benjamin W Priest** and Kevin Gold. Utility discounting explains informational website traffic patterns before a hurricane. In *Proceedings of the 22nd International Conference on World Wide Web*, pages 53–54, 2013
- [38] Kevin Gold, **Benjamin W Priest**, and Kevin M Carter. An expectation maximization approach to detecting compromised remote access accounts. In *The Twenty-Sixth International FLAIRS Conference*, 2013

PEER-REVIEWED
JOURNAL
PUBLICATIONS

- [39] Chinedu Eleh, Yunli Zhang, Rafael Bidese, **Benjamin W Priest**, Amanda L. Muyskens, Roberto Molinari, and Nedret Billor. Stellar blend image classification using computationally efficient Gaussian processes. *Communications in Statistics - Case Studies and Data Analysis*, to appear
- [40] Gregory Sallaberry, Robert E. Armstrong, **Benjamin W Priest**, Michael D. Schneider, and Amanda L. Muyskens. A scalable Gaussian process approach to shear mapping with MuyGPS. *The Astrophysical Journal*, 981(2):166, 2025
- [40] Gregory Sallaberry, Robert E. Armstrong, **Benjamin W Priest**, Michael D. Schneider, and Amanda L. Muyskens. A scalable Gaussian process approach to shear mapping with MuyGPS. *The Astrophysical Journal*, 981(2):166, 2025
- [41] Imène Goumiri, Amanda L. Muyskens, **Benjamin W Priest**, Robert E. Armstrong, and J. Luc Peterson. Uncertainty quantification to detect resident space object anomalies. *The Defense Systems Information Analysis Center (DSIAC) Journal*, 8(2):TBD, 2024

- [42] Juliette Mukangango, Amanda L. Muyskens, and **Benjamin W Priest**. A robust approach to Gaussian process implementation. *Advances in Statistical Climatology, Meteorology and Oceanography*, TBD(TBD):TBD, 2024
- [43] James J Buchanan, Michael D Schneider, Robert E Armstrong, Amanda L Muyskens, **Benjamin W Priest**, and Ryan J Dana. Gaussian process classification for galaxy blend identification in LSST. *The Astrophysical Journal*, 924(2):94, 2022
- [13] Amanda L Muyskens, Imène R Goumiri, **Benjamin W Priest**, Michael D Schneider, Robert E Armstrong, Jason Bernstein, and Ryan Dana. Star–galaxy image separation with computationally efficient Gaussian process classification. *The Astronomical Journal*, 163(4):148, 2022

OPEN-SOURCE SOFTWARE

- [8] Keita Iwabuchi, Grace Li, Trevor Steil, Thomas Stanley, and **Benjamin W Priest**. ClaMS: HPC hierarchical clustering. <https://github.com/LLNL/clams>, 2026. GitHub/LLNL repository
- [7] **Benjamin W Priest**. powersqueeze: A high-performance oblivious embedding toolkit for large, sparse data. <https://github.com/LLNL/powersqueeze>, 2025. GitHub/LLNL repository
- [2] **Benjamin W Priest** and Alec Dunton. krowkee: A high-performance computing toolkit for scalably summarizing many data streams in distributed memory. <https://github.com/LLNL/krowkee>, 2022. GitHub/LLNL repository
- [12] **Benjamin W Priest**, Alec Dunton, Imène Goumiri, and Greg et al. Sallaberry. MuyGPyS: A fast, pure python implementation of the MuyGPs gaussian process realization and training algorithm. <https://github.com/LLNL/MuyGPyS>, 2021. GitHub/LLNL repository
- [10] Trevor Steil, Keita Iwabuchi, Roger Pearce, and **Benjamin W Priest** et al. YGM. <https://github.com/LLNL/ygm>, 2022. GitHub/LLNL repository
- [3] Trevor Steil, Keita Iwabuchi, and **Benjamin W Priest**. SaltAtlas. <https://github.com/LLNL/saltatlas>, 2024. GitHub/LLNL repository

ARXIV PAPERS

- [44] Ukamaka V. Nnyaba, Hewan M. Shemtaga, David W. Collins, Amanda L. Muyskens, **Benjamin W Priest**, and Nedret Billor. Enhancing electrocardiography data classification confidence: A robust gaussian process approach (MuyGPs). *arXiv preprint arXiv:2409.04642*, 2024
- [45] Jason Bernstein, Alec M Dunton, and **Benjamin W Priest**. An analysis of the johnson-lindenstrauss lemma with the bivariate gamma distribution. *arXiv preprint arXiv:2305.17123*, 2023
- [11] Amanda Muyskens, **Benjamin W Priest**, Imène Goumiri, and Michael Schneider. MuyGPs: Scalable Gaussian process hyperparameter estimation using local cross-validation. *arXiv preprint arXiv:2104.14581*, 2021
- [17] Matthew Otten, Imène R Goumiri, **Benjamin W Priest**, George F Chapline, and Michael D Schneider. Quantum machine learning using Gaussian processes with performant quantum kernels. *arXiv preprint arXiv:2004.11280*, 2020
- [14] **Benjamin W Priest**. Degreesketch: Distributed cardinality sketches on massive graphs with applications. *arXiv preprint arXiv:2004.04289*, 2020

TECH
REPORTS

- [46] K Wood, AM Dunton, A Muyskens, and BW Priest. Bayesian hyperparameter optimization in gaussian processes using statistical coverage. Technical report, Lawrence Livermore National Lab.(LLNL), Livermore, CA (United States), 2022
- [47] Andrei Filippov, Imène R Goumiri, and **Benjamin W Priest**. Genetic algorithm for hyperparameter optimization in Gaussian process modeling. Technical report, Lawrence Livermore National Lab.(LLNL), Livermore, CA (United States), 2020

WORKING
PAPERS

- [48] Thomas Stanley, **Benjamin W Priest**, Trevor Steil, and Nate Veldt. Metric forest completion: Parallel and fast. **To Be Submitted To IPDPS 2027**
- [49] Edem Boahen, **Benjamin W Priest**, T. S. Jayram, Grace Li, and Geoffrey Sanders. Approximate multi-matrix multiplication for streaming power iteration clustering. **In preparation**
- [50] **Benjamin W Priest**, Edem Boahen, Keita Iwabuchi, Trevor Steil, and Geoffrey Sanders. Oblivious embeddings for large sparse data via distributed power iteration. **To Be Submitted to IPDPS 2027**
- [51] Keita Iwabuchi, Grace Li, Thomas Stanley, Trevor Steil, Geoffrey Sanders, and **Benjamin W Priest**. Clams: General clustering at massive scale for high-dimensional, non-euclidean data. **In preparation**
- [52] Grace Li, Trevor Steil, **Benjamin W Priest**, and Keita Iwabuchi. Fast clustering of antibody sequences using graph algorithms. **In preparation**

OTHER
CONFERENCE
PUBLICATIONS

- [53] **Benjamin W Priest** and George Cybenko. Approximating centrality in evolving graphs: toward sublinearity. In *Sensors, and Command, Control, Communications, and Intelligence (C3I) Technologies for Homeland Security, Defense, and Law Enforcement Applications XVI*, volume 10184, pages 58–66. SPIE, 2017
- [54] **Benjamin W Priest** and George Cybenko. Efficient inference of hidden markov models from large observation sequences. In *Sensors, and Command, Control, Communications, and Intelligence (C3I) Technologies for Homeland Security, Defense, and Law Enforcement Applications XV*, volume 9825, pages 179–187. SPIE, 2016

BOOK CHAPTERS

- [55] Geoffrey Sanders, Roger Pearce, **Benjamin W Priest**, and Trevor Steil. Massive-scale distributed triangle computation and applications. In *Massive Graph Analytics*, pages 127–158. Chapman and Hall/CRC, 2022
- [56] **Benjamin W Priest**, George Cybenko, Satinder Singh, Massimiliano Albanese, and Peng Liu. Online and scalable adaptive cyber defense. In *Adversarial and Uncertain Reasoning for Adaptive Cyber Defense*, pages 232–261. Springer, 2019

CONFERENCE
TALKS

Benjamin W Priest. Scaling Graph Clustering with Distributed Sketches. At: *2020 High Performance Extreme Computing Conference*, HPEC 2020. Waltham, CA, USA (virtual conference), 21–25 September 2020.

Benjamin W Priest. Approximating centrality in evolving graphs: toward sublinearity. At: *2017 SPIE Defense + Security Conference*, SPIE D+S. Anaheim, CA, USA, 9–13 April 2017.

Benjamin W Priest. Efficient Inference of hidden Markov models from large observations sequences. At: *2016 SPIE Defense + Security Conference*, SPIE D+S. Anaheim, CA, USA, 17–21 April 2016.

Benjamin W Priest. Agent-based simulation in support of moving target cyber defense technology development and evaluation. At: *18th Symposium on Communications & Networking, 2015 ACM Spring Simulation Multi-Conference*, CNS/SpringSim. Alexandria, VA, USA, 12–15 April 2015.

Benjamin W Priest. Characterizing latent user interests on enterprise networks. At: *2014 International Florida Artificial Intelligence Research Society Conference*, FLAIRS. Pensacola Beach, FL, USA, 21–23 May 2014.

INVITED TALKS

Benjamin W Priest. High-fidelity enterprise network emulation using the GOSMR architecture. In: *2014 MIT Lincoln Laboratory Cyber and Net-Centric Workshop*, CNW. June, 2014.

POSTER
PRESENTATIONS

Benjamin W Priest. Estimating edge-local triangle count heavy hitters in edge-linear time and almost-vertex-linear space. At: *GraphChallenge Workshop at the IEEE High Performance Extreme Computing Conference*, HPEC. 25–27 September 2018.

Benjamin W Priest. Efficient Sublinear Estimation of Local Triangle Count Heavy Hitters. At: *2018 Summer Student Poster Symposium at Lawrence Livermore National Laboratory*. 9 August 2018.

Benjamin W Priest. Characterization of latent social networks discovered through computer network logs. At: *Networks in the Social and Information Sciences workshop of the 29th Annual Conference on Neural Information Processing Systems*, NIPS. Montreal, Canada, 12 December 2015.

Benjamin W Priest. Utility discounting explains informational website traffic patterns before a hurricane. At: *22nd International World Wide Web Conference*, WWW. 2013. Rio de Janeiro, Brazil, 13–17 May 2013.

PROJECT
PROPOSALS

Awaiting

- PI**, [Title Omitted],
- LLNL Strategic Partnership Program.
 - \$1,000,000/yr.
 - FY 28-29.

Awarded

- PI**, [Title Omitted],
- LLNL Strategic Partnership Program.
 - \$500,000/yr.
 - May 1, 2026 to April 30, 2027.

Co-PI, “RAMSHORN: Randomized Algorithms for Massive, Scientific, Higher-Order Relational Networks”,

- DOE ASCR DE-FOA-0003432,
- \$1,000,000/yr (\$350,000/yr for LLNL).
- October 1, 2025 to September 30, 2028.

Co-I, “Scalable Bayesian Optimization”,

- LLNL LDRD 26-FS-XXX,

- \$150,000/yr.
- October 1, 2025 to September 30, 2026.

Co-I, “FOCUS: Framework for Optimal Climate-Infrastructure Solutions and Decision Support”,

- LLNL LDRD 25-SI-XXX,
- \$3,450,000/yr.
- October 1, 2024 to September 30, 2027.

PI, “Hierarchical Graph-Based Clustering in Distributed Memory”,

- LDRD 24-ERD-024
- \$600,000/yr.
- October 1, 2023 to September 30, 2026.

Co-I, “HPC-Enabled Detection System for Petabyte Scale Astronomy Surveys”,

- LLNL LDRD 23-ERD-028
- \$700,000/yr.
- October 1, 2022 to September 30, 2025.

Co-I, “MuyGPs: Non-Stationary Gaussian Processes at HPC Scales”,

- LLNL LDRD 22-ERD-028
- \$850,000/yr.
- October 1, 2021 to September 30, 2024.

Co-I, “Scalable Uncertainty Quantification Using Gaussian Processes Surrogate Models”,

- LLNL LDRD 21-FS-037
- \$100,000/yr.
- January 1, 2021 to September 30, 2021.

Co-I, “Interactive Exploratory Graph-Enabled Data Analytics at HPC Scales”,

- LLNL LDRD 21-ERD-020
- \$500,000/yr.
- October 1, 2020 to September 30, 2023.

PI, “Scalable Approximate Graph Clustering”,

- LLNL LDRD 20-ERD-037
- \$150,000/yr.
- February 1, 2020 to September 30, 2020.

Declined

co-PI, [Title Omitted],

- LLNL Strategic Partnership Program.
- 2.5 FTEs/yr.
- FY 26-28.

PI, “Efficient Data Reduction via Active Learning Augmented Coresets”,

- DOE ASCR DE-FOA-0003266
- \$1,000,000/yr.
- FY 25-27.

Co-I, “Scalable Physics-Constrained Emulation of Stochastic Moments of Quantum Processing Units”,

- DOE ASCR DE-FOA-0003300
- \$500,000/yr.
- FY 25-26.

Co-I, “Science with LSST Year 1 Data”,

- LLNL LDRD ER
- \$1,000,000/yr.
- FY 25-27.

Co-I, “Decentralized Strategic Decision Making via Linearly Solvable Markov Games”,

- LLNL LDRD ER
- \$600,000/yr.
- FY 25-27.

Co-I, “Uncertainty-Aware Realtime Algorithmic Calibration in Distributed Computing Environments”,

- DOE ASCR DE-FOA-0003266
- \$1,200,000/yr.
- FY 24-27.

PI, “Scalable and Highly Concurrent Sampling Algorithms for Scale-Free Graphs”,

- DOE ASCR DE-FOA-0002722
- \$800,000/yr.
- FY 23-25.

Co-I, “Probabilistic AI Pipeline Modules for Rubin LSST Dark Energy”,

- DOE ASCR DE-FOA-0002705
- \$1,000,000/yr.
- FY 23-25.

Co-PI, “Scalable Single-Pass Compressive Autoencoders”,

- LLNL LDRD Feasibility Study
- \$150,000/yr.
- FY 23.

Co-I, “EpochGrafts: Relational Data Fusion via Dynamic Graph Analysis”,

- LLNL LDRD ER
- \$500,000/yr.
- FY 22-24.

PI, “Scalable Non-stationary Approximate Gaussian Processes”,

- DOE ASCR DE-FOA-0002493
- \$800,000/yr.
- FY 22-24.

PI, “Distributed Memory Sketching Algorithms at HPC Scales”,

- DOE ASCR DE-FOA-0002497
- \$400,000/yr.
- FY 22-23.

PI, “*croquis*: Distributed Subspace Embeddings for High Performance Computing”,

- LLNL ISCP Tech Base
- \$100,000/yr.
- FY 21-22.

MENTORSHIP

Postdocs

Grace Li, Postdoc in Applied Mathematics.

- Lawrence Livermore National Laboratory
- Distributed memory graph algorithms, clustering, and analysis.
- Exascale energy grid expansion modeling under topological uncertainty.
- 2024-current.

Luke Erlandson, Postdoc in Applied Mathematics.

- Lawrence Livermore National Laboratory
- Hierarchical Approximation of Gaussian process kernels.
- 2023.

Alec Dunton, Postdoc in Applied Mathematics.

- Lawrence Livermore National Laboratory
- Fast and scalable Gaussian process approximation in distributed memory.
- 2021-2023.

Post-Bacs and Post-Masters

Gregory Sallaberry, Post Master’s in Astronomy.

- Lawrence Livermore National Laboratory.
- Modeling billion-scale cosmic lensing shear models with scalable Gaussian processes.
- 2023-2024.

Keegan Kresge, Post Baccalaureate in Mathematics.

- DOD.
- K-Nearest Neighbors performance of exponentiated subspace embeddings on large graphs.
- Summer 2023.

Students

Alima Subedi, PhD student in Computer Science.

- Missouri University of Science and Technology.
- Near-duplicate filtering and data partitioning via locality sensitive hashing to accelerate distributed k nearest neighbor graph construction.
- Primary Advisor: Satish Puri.
- Summer 2026-present.

S M Shovan, PhD student in Computer Science.

- Missouri University of Science and Technology.
- Accelerating k nearest neighbor graph construction GPU kernels via matrix multiplication cores.
- Primary Advisor: Sajal K. Das.
- Summer 2026-present.

Omkar Bhalerao, PhD student in Computer Science.

- UC Santa Clara.
- Approximate Near Shortest Path Counting and temporal clustering coefficient estimation.
- Primary Advisor: C. Seshadhri.
- Summer 2025-present.

Jacob Horsley, PhD student in Mathematics.

- UC San Diego.
- Clustering Graphs using Additive Spectral Decompositions.
- Primary Advisor: Geoffrey Sanders.
- Summer 2025.

Thomas Stanley, PhD student in Computer Science.

- Texas A & M University.
- Metric minimum spanning tree estimation via serial and distributed metric forest completion.
- Primary Advisor: Nate Veldt.
- Autumn 2024-present.

Edem Boahen, PhD student in Mathematics.

- Michigan State University.
- Clustering guarantees for power iteration algorithms on degree-corrected stochastic block models.
- Primary Advisor: Mark Iwen.
- Summer 2024-present.

Sunanda Thirunabukkarasu (Mila Arasu), BS student in Astrophysics.

- Arizona State University.
- Simulating cosmic lensing shear with realistic B- and E-mode shifts.
- Summer 2024.

Lance Fletcher, PhD student in Computer Science.

- Texas A & M University.
- Massively parallel random walk sampling and embeddings.
- Primary Advisor: Roger Pearce.
- Autumn 2023-present.

Juliette Mukangango, PhD student in Statistics.

- Colorado School of Mines.
- Novel loss and objective functions for outlier robustness in training sparse MuYGPs models.

- Primary Advisor: Douglas Nychka.
- Summer 2023.

Akil Andrews, PhD student in Computer Science.

- University of New Mexico.
- Adaptive Bayesian optimization under changing data representations.
- Primary Advisor: Melanie E Moses.
- Summer 2023.

Ian McGovern, PhD student in Statistics.

- University of California, Los Angeles.
- Uncertainty analysis of hybrid deep neural network and Gaussian process predictions.
- Primary Advisor: Frederic Schoenberg.
- Summer 2023.

Abiodun Sumonu, PhD student in Mathematics.

- University of Alabama.
- Survey of Biclustering Algorithms.
- Summer 2023.

Marina Dunn, Masters student in Data Science,

- University of California, Riverside.
- Visualizing sparse Gaussian process optimization.
- Summer 2022.

Killian Wood, PhD Student in Applied Mathematics.

- University of Colorado, Boulder.
- Multiscale Bayesian optimization of MuiGPs.
- Primary Advisor: Emiliano Dall’Anese.
- Summer 2022.

Michał Lisicki, PhD student in Computer Science.

- University of Guelph.
- Distributional reinforcement learning on gridworld environments.
- Primary Advisor: Graham Taylor.
- Summer 2022.

Sudharshan Srinivasan, PhD student in Computer Science.

- University of Oregon.
- Communication optimization for highly non-uniform distributed graph algorithms.
- Primary Advisor: Boyana Norris.
- Summer 2021.

Alec Dunton, PhD student in Applied Mathematics.

- University of Colorado, Boulder.
- Parameter sensitivity of stochastic block models under subspace embeddings.
- Primary Advisor: Alireza Doostan.
- Summer 2020.

Teams and Challenges

Davy Walker, Ukamaka Nnyaba, and Hewan Shemtaga,

- Master’s and PhD Students in Computer Science, Auburn University.
- Auburn Data Science Capstone Project.
- ECG Time Series Classification Using Computationally Efficient Gaussian Processes.
- Autumn 2023.

Rafael Bidese, Chinedu Eleh, and Yunli Zhang,

- Master’s and PhD Students in Computer Science, Auburn University.
- Auburn Data Science Capstone Project.
- Stellar Blend Image Classification Using Computationally Efficient Gaussian Processes.
- Autumn 2022.

Jocelyn Ornelas, Alan Triano, Cristian Espinosa, Denilson Fuentes, and Rahul Ravi,

- A PhD Student (Jocelyn) and 4 undergraduates in Data Science programs.

- University of California, Merced.
- LLNL Data Science Challenge.
- Asteroid detection and orbit extraction from astronomy corpora.
- Sprint 2021.

PROFESSIONAL SERVICE

Conference Service

- Program Committee: The Fourteenth International Conference on Learning Representations (ICLR26). Rio de Janeiro, Brazil. April 23-27, 2026.
- Program Committee: SIAM International Conference on Data Mining (SDM24). Houston, Texas, USA. April 18-20, 2024.
- Program Committee: 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD23). Long Beach, California, USA. August 6-10, 2024.
- Program Committee: 28th International AAAI Florida Artificial Intelligence Research Symposium Conference, FLAIRS-28. Hollywood, Florida, USA. May 18-20, 2015.

Journal Service

- Reviewer: American Astronomical Society: The Astrophysical Journal. 2022.

Committee Service

- Lead/Director. LLNL/Data Science Institute – Data Science Summer Institute. 2024-present.
- Committee Member. LLNL/Laboratory Directed Research and Development – Exploratory Research Project Review Committee. 2026-present.
- Committee Member. LLNL/Computing Advisory Committee. Subcommittees: “Thrive Conversations”, “Improving Student Pipelines”, and “Stress and Burnout Initiative”. 2023-present.
- Leader/Organizer. LLNL/Computing Summer SLAM! presentation session for summer students. Summer, 2023, 2024, 2025, & 2026.
- Committee Member. LLNL/Computing Strategic Initiative for DEI in Recruiting, Outreach, and Hiring. 2021.

Mentorship Service

- Instructor: “DevOps for Data Scientists”. LLNL Data Science Summer Institute. Summer, 2023, 2024, & 2025.
- Team Mentor. Auburn University Data Science Capstone project. “Enhancing Electrocardiography Data Classification Confidence: A Robust Gaussian Process Approach” Autumn, 2023.
- Team Mentor. Auburn University Data Science Capstone project. “Stellar Blend Image Classification Using Computationally Efficient Gaussian Processes”, Autumn, 2022.
- Instructor: “The Streaming Data Model and Selected Algorithms”. LLNL Data Science Summer Institute. Summer, 2022.
- Team Coach. LLNL/UC Merced Data Science Challenge. Summer, 2022.

AWARDS

LLNL SPOT Awards

- 2025: Championing of academic partnership programs within LLNL/Computing complex.
- 2024: Excellence in instructing and mentoring students as part of the Data Science Summer Institute (DSSI).
- 2023: Outstanding leadership of the Summer 2023 Summer SLAM! student presentation seminar.
- 2022: Contributions to COMP Strategic Initiative for DEI in recruiting, outreach, and hiring.
- 2021: Excellence in mentorship as a part of the LLNL/UC Merced Data Science Challenge.

LLNL Deputy Director Science & Technology Excellence in Publication Award

- 2023: Gaussian process classification for galaxy blend identification in LSST
- 2023: Light curve completion and forecasting using fast and scalable Gaussian processes (MuyGPs)
- 2021: Scaling Graph Clustering with Distributed Sketches

Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference
2022: Best Machine Learning Paper

High Performance Extreme Computing (HPEC) Graph Challenge
2020: Graph Challenge Champion
2019: Graph Challenge Champion

International Conference on Security and Cryptography (SECRYPT)
2018: Best Paper Award

MIT Lincoln Laboratory
2015: Lincoln Scholar Fellowship

The Ohio State University
2010: Phi Beta Kappa Inductee
2010: Bingham Award in Philosophy
2008: Kenneth Cummings Scholarship
2007-2011: Distinguished Merit Scholarship
2007-2011: Ohio Academic Scholarship

TECHNICAL EXPERTISE

Mathematics

Applied Mathematics
Real Analysis
Measure Theory
Graph Theory
Combinatorics

Computer Science and Engineering

Distributed & parallel algorithms
Streaming algorithms & sketching
Data structures

Data Science and Processing

Probability & Random Variables
Statistics & Estimation
Machine learning & deep learning
Numerical Optimization
Stochastic Processes
Information Theory
Communication Theory
Large Language Models

Programming and Scripting Languages

C/C++, Python, Bash,
Julia, Java, R, MATLAB

Distributed Computing

MPI, Hadoop MapReduce, Lustre

Analytical Software

Keras, TensorFlow, PyTorch, Mathematica

Utility Software

Git, GitHub/Gitlab/Bitbucket
L^AT_EX, B_IB_TE_X
Microsoft, LibreOffice, Google Suite
Agentic frameworks, e.g., codex

Operating Systems

Linux, RedHat, and other UNIX variants
Apple OS X

Interpersonal

Teamwork and communication
Leadership and mentoring
Public and technical speaking

TEACHING EXPERIENCE

Thayer School of Engineering at Dartmouth College, Hanover, VT, USA

Teaching Assistant

Instructor for ENGS/QBS 108: Applied Machine Learning **Autumn 2017**

- Collaborated with instructors to develop course curriculum aimed at graduate engineering and computer science students and taught $\sim 25\%$ of course lecture content.
- Led team of 4 teaching assistants
- Provided group and one-on-one assistance to students covering lecture topics
- Planned, wrote, and graded all student assignments

Instructor for ENGS 177: Decision Making Under Risk and Uncertainty **Winter 2017**

- Planned and taught a weekly recitation covering practical machine learning topics
- Provided ground and one-on-one assistance to students covering lecture topics
- Wrote student assignments with the assistance of the instructor and provided grading

The Ohio State University, Columbus, OH, USA

Teaching Assistant

Instructor for CSE 625: Automata and Formal Languages **Summer & Autumn 2010**

- Planned and taught a weekly recitation covering details and proofs of lecture topics

- Graded student assignments
- Grader for CSE 560: System Software Design and Development
- Graded student assignments and held office hours

Summer 2010

SECURITY
CLEARANCE
CITIZENSHIP

DOE Q-clearance (Spring 2024) with SCI (Autumn 2024).

USA